

Combinatorial Purged CV

CPCV is used to test strategies without overfitting or relying on single backtest path.

How it works?

1. Partitioning data

- T observations are split into N groups ($G_1, G_2, \dots, G_N$)
- Choose among N groups, k testing groups and generate all combinations

2. Counting split and backtest paths

- Total number of unique ways to select k from N is:

$$\binom{N}{k} = \binom{N}{N-k} = \frac{N!}{k!(N-k)!}$$

- For a complete backtest path you need all groups.

Defining $\varphi(N, k)$ as amount of backtest paths:

$$\varphi(N, k) = \frac{k}{N} \times \binom{N}{k}$$

3. Training and testing

- For each split train on N-k groups and test on k groups with purge + embargo
- Recreate $\varphi(N, k)$ backtest paths instead of one (ex: WF backtest)
- Maximise training ratio Θ while keeping φ high.

$$\Theta = \frac{\text{Training groups}}{\text{Total groups}} = \frac{N-k}{N} = 1 - \frac{k}{N}, \text{ if } \Theta = 80\% \text{ move training data for model}$$

- Bound $k \leq N/2$ ensure that $\Theta \geq 1 - \frac{N/2}{N} \geq 50\%$ of Training
- Result is a distribution of performance on φ backtest paths.

Example with $N=6, k=2, \binom{6}{2}=15$ splits, $\varphi(6,2)=5$ backtest paths

	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8	S_9	S_{10}	S_{11}	S_{12}	S_{13}	S_{14}	S_{15}
G_1	x	x	x	x	x										
G_2	x					x	x	x	x						
G_3		x				x				x	x	x			
G_4			x				x			x			x	x	
G_5				x				x			x		x		x
G_6					x				x			x		x	x

x : Testing groups

	P_1	P_2	P_3	P_4	P_5
G_1	1	2	3	4	5
G_2	1	6	7	8	9
G_3	2	6	10	11	12
G_4	3	7	10	13	14
G_5	4	8	11	13	15
G_6	5	9	12	14	15

Splits to fetch test from

Reducing performance variance

Assuming we test I strategy variations on martingale where $E[y_i] = 0$ (sharp ratio)

Performance can be decomposed into standard normal noise $z_i \sim N(0,1)$ multiplied by estimator deviation $\sigma(y_i)$:

$$y_i = z_i \cdot \sigma(y_i)$$

Researchers pick best strategy: $\max_i y_i$, therefore with Extreme Value Theory we get:

$$E[\max_i y_i] \approx (1-\gamma) Z^{-2} \left[1 - \frac{1}{I} \right] + \gamma Z^{-2} \left[1 - \frac{1}{I} e^{-1} \right] \leq \sqrt{2 \log(I)}$$

$$E[\max_i y_i] \leq \sqrt{2 \log(I)} \cdot \sigma(y_i) \quad (\sigma(y_i) \text{ linearly inflates performance})$$

In CPCV we take average of performances: $\mu_i = \frac{1}{\varphi} \sum_j y_{i,j}$

$$\text{Therefore, } \sigma^2(\mu_i) = \text{Var}\left(\frac{1}{\varphi} \sum_j y_{i,j}\right) = \frac{1}{\varphi^2} \left[\sum_j \text{Var}(y_{i,j}) + \sum_{j \neq k} \text{Cov}(y_{i,j}, y_{i,k}) \right]$$

$$= \varphi^{-2} \sigma_i^2 (1 + (\varphi-1) \bar{\rho}_i)$$

Because $\bar{\rho}_i < 1$, low correlation between paths, then $\varphi^{-2} \sigma_i^2 \leq \sigma^2(\mu_i) < \sigma_i^2$

Plugging in $E[\max_i y_i] \leq \sqrt{2 \log(I)} \cdot \sigma(\mu_i)$ ($\sigma(\mu_i)$ drastically smaller), so less chance of false discoveries.